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Machine learning AAT Report on

"Urban Air Quality Forecasting and Health Advisory System using Machine Learning"

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B.M.S. COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Machine Learning AAT titled "Urban Air Quality Forecasting and Health Advisory System using Machine Learning" has been carried out by Mohammed Kalandar (1BM23CS195), Milan Paudel (1BM23CS191), Manojkumar (1BM23CS185), Makadia Rishit Dilipbhai (1BM23CS177) during the academic year 2025-2026.

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B.M.S. COLLEGE OF ENGINEERING

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DECLARATION

We, Mohammed Kalandar (1BM23CS195), Milan Paudel (1BM23CS191), Manojkumar (1BM23CS185, Makadia Rishit Dilipbhai (1BM23CS177), students of 6^{6h} Semester, B.E, Department of Computer Science and Engineering, B.M.S. College of Engineering, Bangalore, hereby declare that, this assignment work entitled **"Urban Air Quality Forecasting and Health Advisory System using Machine Learning"** has been carried out by us under the guidance of **Dr. Pallavi G B,** Professor, Department of CSE, B.M.S. College of Engineering, Bangalore during the academic semester Jan-May 2026. We also declare that to the best of our knowledge and belief, the assignment reported here is not from part of any other report by any other students.

Signature of the Candidates

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Urban Air Quality Forecasting and Health Advisory System using Machine Learning

Problem statement

Air pollution in urban areas poses severe risks to public health, contributing to respiratory and cardiovascular diseases. Traditional monitoring systems provide only reactive insights and lack predictive capabilities. This project aims to design a machine learning-based system capable of forecasting Air Quality Index (AQI) values and categorizing them into health-relevant classes. The system leverages historical air quality data across multiple Indian cities to generate predictive insights and support proactive health advisories.

1. Problem Formulation

The problem is formulated as a dual supervised learning task, combining regression and classification to achieve both precision and interpretability.

1.1 Task Definition

- Regression → Predict continuous AQI value
- Classification → Predict AQI category (Good, Moderate, etc.)

1.2 Input and Output Structure

Input (X):

- Pollutants: PM2.5, PM10, NO_2, CO, SO_2, etc.
- Time: Year, Month, Day
- Location: City (encoded)

Output (Y):

- AQI (numerical)
- AQI_Bucket (categorical)

1.3 Key Assumptions

- Historical pollution trends reflect future conditions
- Missing values can be handled using statistical methods
- City encoding captures spatial variation

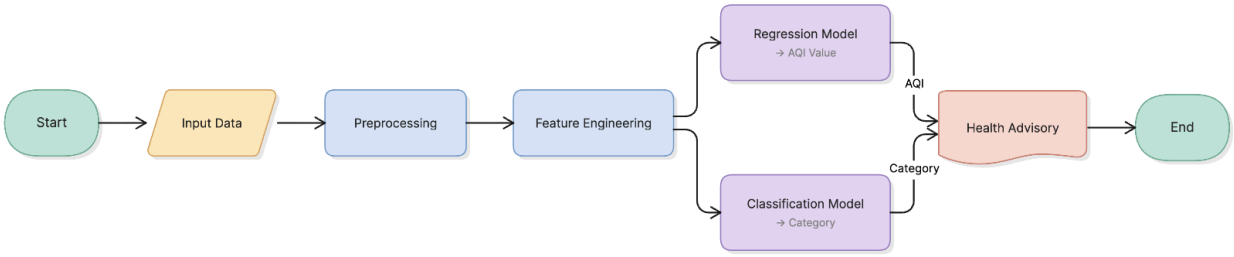

Fig 1.1 System Flow

2. Dataset Strategy

The system uses the **Air Quality Data in India Kaggle** dataset, which provides multi-city air quality data across India.

The dataset is suitable because it captures temporal trends, spatial diversity, and pollutant variations, making it ideal for real-world modeling.

Key Challenges

- Missing values in multiple pollutant columns
- Noisy real-world data
- Inconsistent entries

Handling Strategy

- Invalid dates → removed using safe parsing
- Missing AQI → dropped
- High-missing feature (Xylene) → removed
- Remaining missing values → median imputation

```
df = pd.read_csv("city_day.csv")
df.head()
```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	NaN
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	NaN	NaN
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	NaN	NaN

3. Feature Engineering

Feature engineering transforms raw data into meaningful inputs that improve model performance.

Temporal features such as year, month, and day are extracted to capture seasonal pollution patterns, while city information is encoded to represent spatial variation.

Engineered Features

- Year, Month, Day (from Date)
- One-hot encoding for City
- Retained original city column for evaluation

Why It Matters

- Captures seasonality (winter pollution spikes)
- Captures regional pollution differences
- Improves model learning capability

```
# Extract time features
# Save original city for later evaluation
df['City_Original'] = df['City']

# Time features
df['year'] = df['Date'].dt.year
df['month'] = df['Date'].dt.month
df['day'] = df['Date'].dt.day

# Encode city
df = pd.get_dummies(df, columns=['City'], drop_first=True)

df.head()
```

1 to 5 of 5 entries Filter

index	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	AQI	AQI_Bucket	City_Original	year	month	day	City_Aizawl	City_Amaravati
28	2015-01-29 00:00:00	83.13	96.18	6.93	28.71	33.72	16.31	6.93	49.52	59.76	0.02	0.0	209.0	Poor	Ahmedabad	2015	1	29	false	false
29	2015-01-30 00:00:00	79.84	96.18	13.85	28.68	41.08	16.31	13.85	48.49	97.07	0.04	0.0	328.0	Very Poor	Ahmedabad	2015	1	30	false	false
30	2015-01-31 00:00:00	94.52	96.18	24.39	32.66	52.61	16.31	24.39	67.39	111.33	0.24	0.01	514.0	Severe	Ahmedabad	2015	1	31	false	false
31	2015-02-01 00:00:00	135.99	96.18	43.48	42.08	84.57	16.31	43.48	75.23	102.7	0.4	0.04	782.0	Severe	Ahmedabad	2015	2	1	false	false
32	2015-02-02 00:00:00	178.33	96.18	54.56	35.31	72.8	16.31	54.56	55.04	107.38	0.46	0.06	914.0	Severe	Ahmedabad	2015	2	2	false	false

4. Model Design Strategy

The system adopts a hybrid modeling approach combining regression and classification to address both prediction accuracy and interpretability. The goal is not just to estimate AQI values but also to translate them into actionable categories for users.

Model Choices

- Regression Model: Random Forest Regressor
- Classification Model: Logistic Regression

4.1 Random Forest (Regression)

Random Forest is chosen because AQI prediction involves complex, non-linear interactions between multiple pollutants and temporal factors.

Key reasons:

- Captures non-linear relationships (e.g., PM2.5 + humidity effects)
- Works well with tabular structured data
- Robust to noise and missing values
- Reduces overfitting through ensemble averaging (bagging)

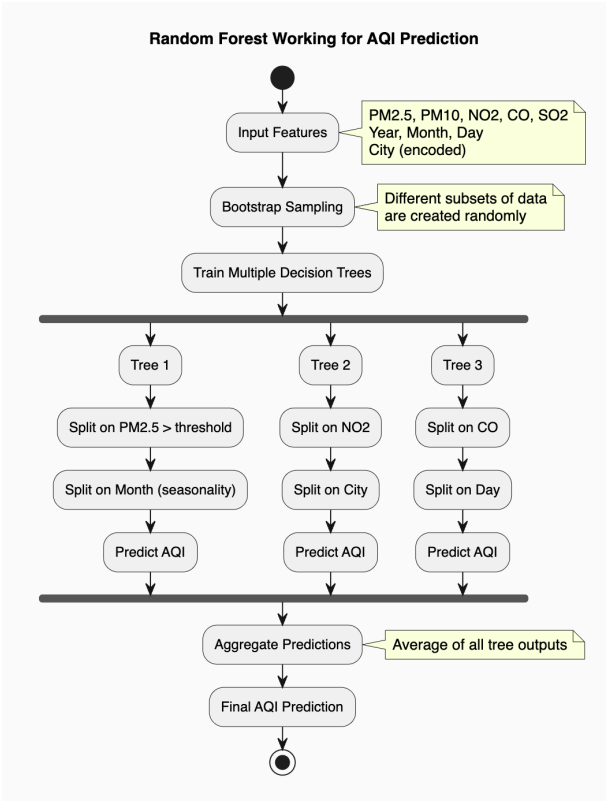

Fig 4.1 Random Forest Working Diagram

Code:

```
model = RandomForestRegressor(n_estimators=100)

model.fit(X_train, y_train)

predictions = model.predict(X_test)

results = pd.DataFrame({

    "City": city_test.values,

    "Actual AQI": y_test.values,

    "Predicted AQI": predictions

})

print(results.head(10))
```

Output:

```
                   City        RMSE
25              Aizawl    6.407845
14           Ernakulam    7.373175
5           Coimbatore    8.086251
1               Mumbai   11.138338
20            Shillong   11.145767
22  Thiruvananthapuram   12.984711
23           Amaravati   13.693417
18              Bhopal   14.859567
8              Kolkata   18.658294
16       Visakhapatnam   19.528618
6           Chandigarh   20.598000
24              Jaipur   22.143347
17               Kochi   22.728474
10           Hyderabad   24.330390
4            Bengaluru   28.984073
```

4.2 Logistic Regression (Classification)

Logistic Regression is used as a baseline classification model to map AQI into categories.

Key reasons:

- Simple and computationally efficient
- Provides interpretable decision boundaries

- Suitable for multi-class classification
- Easy to deploy in real-time systems

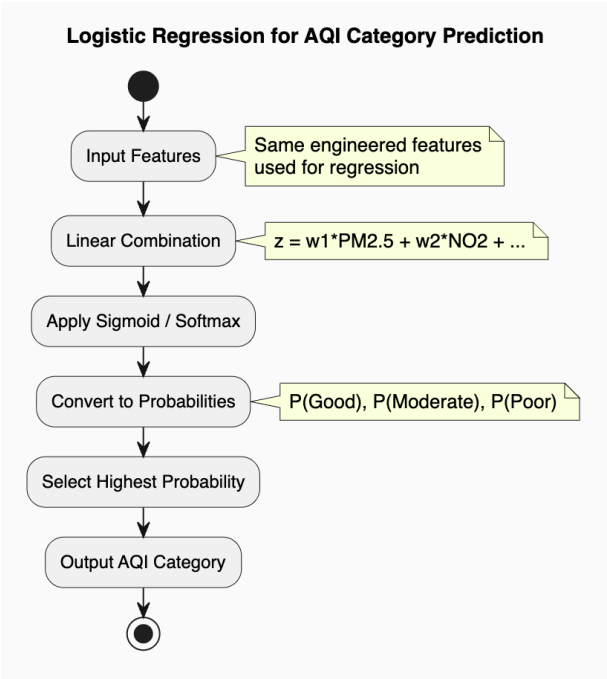

Fig 4.2 Logistic Regression Working Diagram

Code:

```
clf = LogisticRegression(max_iter=1000)

clf.fit(X_train_cls, y_train_cls)

category_preds = clf.predict(X_test_cls)

results_cls = pd.DataFrame({

    "City": city_test_cls.values,

    "Actual": y_test_cls.values,

    "Predicted": category_preds

})

results_cls['Actual Label'] = results_cls['Actual'].map(mapping)

results_cls['Predicted Label'] = results_cls['Predicted'].map(mapping)

print(results_cls.head(10))
```

Output:

	City	Actual	Predicted	Actual Label	Predicted Label
0	Chennai	1	1	Moderate	Moderate
1	Mumbai	2	1	Poor	Moderate
2	Amritsar	1	1	Moderate	Moderate
3	Lucknow	1	1	Moderate	Moderate
4	Bengaluru	3	3	Satisfactory	Satisfactory
5	Coimbatore	3	3	Satisfactory	Satisfactory
6	Chandigarh	3	3	Satisfactory	Satisfactory
7	Bengaluru	3	3	Satisfactory	Satisfactory
8	Chennai	3	1	Satisfactory	Moderate
9	Patna	3	3	Satisfactory	Satisfactory

4.3 Training Strategy

- Dataset split into 80% training, 20% testing
- Feature scaling applied for model stability
- Single global model trained using all cities
- Evaluation performed:vGlobally, City-wise (important enhancement)

5. Evaluation Strategy

The evaluation framework is designed to reflect real-world performance requirements.

Metrics Used

- RMSE → Measures AQI prediction error
- Accuracy → Measures category prediction

```
city_rmse = {}

for city in results['City'].unique():
    city_data = results[results['City'] == city]

    if len(city_data) > 10:  # avoid very small samples
        rmse_city = np.sqrt(mean_squared_error(
            city_data['Actual AQI'],
            city_data['Predicted AQI']
        ))
        city_rmse[city] = rmse_city

city_rmse_df = pd.DataFrame(list(city_rmse.items()), columns=['City', 'RMSE'])
city_rmse_df = city_rmse_df.sort_values(by='RMSE')

print(city_rmse_df)
```

```
                City        RMSE
25             Aizawl    6.407845
14          Ernakulam    7.373175
5           Coimbatore   8.086251
1              Mumbai   11.138338
20            Shillong  11.145767
22  Thiruvananthapuram  12.984711
23          Amaravati   13.693417
18             Bhopal   14.859567
8             Kolkata   18.658294
16       Visakhapatnam  19.528618
6           Chandigarh  20.598000
24             Jaipur   22.143347
17              Kochi   22.728474
10           Hyderabad  24.330390
4            Bengaluru  28.984073
```

Validation Strategy

- Train-Test Split (80-20)
- No data leakage ensured

6. Deployment Design

The proposed system is designed as a lightweight, scalable prediction service that delivers AQI forecasts and health categories to end users in real time.

The deployment follows a client–server architecture, where users interact through a frontend interface, and all processing is handled by a backend service hosting the trained machine learning models.

System Workflow

1. User provides input (pollutant values, date, city)
2. Backend performs preprocessing (cleaning, feature engineering, scaling)
3. Random Forest model predicts AQI value
4. Logistic Regression model predicts AQI category
5. System generates output (AQI + category + advisory)
6. Results are returned to the user interface

7. Monitoring and Maintenance Strategy

To ensure the system remains reliable and effective over time, a continuous monitoring and maintenance framework is essential. Since air pollution patterns are dynamic and influenced by seasonal and regional factors, model performance can degrade if not regularly evaluated and updated.

The system tracks performance using key metrics such as RMSE for AQI prediction and accuracy for classification. In addition to overall performance, city-wise evaluation is monitored to identify regional variations and detect if the model underperforms in specific locations.

Monitoring Components

- RMSE tracking over time to detect prediction drift
- Accuracy tracking for classification performance
- City-wise performance monitoring to capture spatial inconsistencies

When performance drops below acceptable levels, corrective actions are triggered. This ensures that the system adapts to changing environmental conditions.

Maintenance Strategy

- Periodic retraining using updated data
- Continuous dataset updates to reflect new pollution patterns
- Logging of predictions and model outputs for analysis

Alerts and Actions

- Automated alerts when performance degrades
- Identification of cities requiring model improvement
- Fallback to previous stable model if needed

8. Ethical, Social, and Risk Considerations

Given that the system directly impacts public health decisions, ethical considerations play a critical role in its design and deployment. Incorrect predictions may lead to harmful consequences, especially for sensitive groups such as children, elderly individuals, and people with respiratory conditions.

The system must therefore be designed with caution, ensuring that predictions are used as guidance rather than absolute decisions.

Risks

- Underestimation of AQI may expose users to harmful conditions
- Overestimation may cause unnecessary panic or restrictions
- Data imbalance across cities may lead to biased predictions

Ethical Measures

- Ensure transparency in model outputs and limitations
- Avoid presenting predictions with overconfidence
- Clearly communicate that the system provides advisory insights

Social Impact

- Raises public awareness about air quality
- Helps individuals make informed health decisions
- Supports vulnerable populations in reducing exposure risk

9. SDG Mapping

SDG 3 — Good Health and Well-being

The system predicts AQI levels and provides health-based categories, enabling users to take preventive actions.
This helps reduce exposure to harmful air, especially for vulnerable populations.

SDG 11 — Sustainable Cities and Communities

City-wise AQI prediction allows identification of pollution patterns across urban areas.
This supports better planning and targeted interventions for cleaner and sustainable cities.

SDG 13 — Climate Action

The system enables continuous monitoring and forecasting of air pollution trends.
This supports early awareness and proactive response to environmental and climate-related issues.